In this chapter, we will be discussing the concepts mainly as continuation of previous classes and discuss most important concepts like Similar Triangles, Basic Proportionality Theorem, Angle Bisector Theorem, the most prominent and widely acclaimed Pythagoras Theorem and much more. Ceva's Theorem and Menelaus Theorem is introduced for the first time. These two new theorems generalize all concurrent theorems that we know. Overall, the study of Geometry will create interest in the deep understanding of objects around us.

Geometry plays vital role in the field of Science, Engineering and Architecture. We see many Geometrical patterns in nature. We are familiar with triangles and many of their properties from earlier classes.

4.2 Similarity



Two figures are said to be similar if every aspect of one figure is proportional to other figure. For example:

The above houses look the same but different in size. Both the mobile phones are the same but they vary in their sizes. Therefore, mathematically we say that two objects are similar if they are of same shape but not necessarily they need to have the same size. The ratio of the corresponding measurements of two similar objects must be proportional.



Fig. 4.1

Here is a box of geometrical shapes. Collect the similar objects and list out.

In this chapter, we will be discussing specifically the use of similar triangles which is of utmost importance where if it is beyond our reach to physically measure the distance and height with simple measuring instruments. The concept of similarity is widely used in the fields of engineering, architecture and construction.



Fig. 4.2

Here are few applications of similarity

(i) By analyzing the shadows that make triangles, we can determine the actual height of the objects.

(ii) Used in aerial photography to determine the distance from sky to a particular location on the ground.

(iii) Used in Architecture to aid in design of their work.

4.2.1 Similar triangles

In class IX, we have studied congruent triangles. We can say that two geometrical figures are congruent, if they have same size and shape. But, here we shall study about geometrical figures which have same shape but proportional sizes. These figures are called "similar".



Fig. 4.3

10th_Maths_Chapter 4_English.indd 162 12/27/2024 3:10:51 PM